

Tutorial 2:

Backpropagation

Administrative: Assignment 1

Due 1/23 11:59pm

- K-Nearest Neighbor
- Linear classifiers: SVM, Softmax

Administrative: EdStem

Please make sure to check and read all pinned EdStem posts.

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^n (W \cdot x)_i^2$

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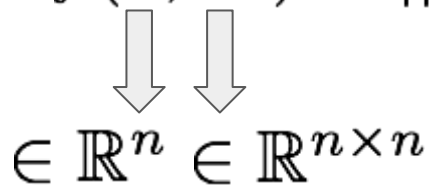
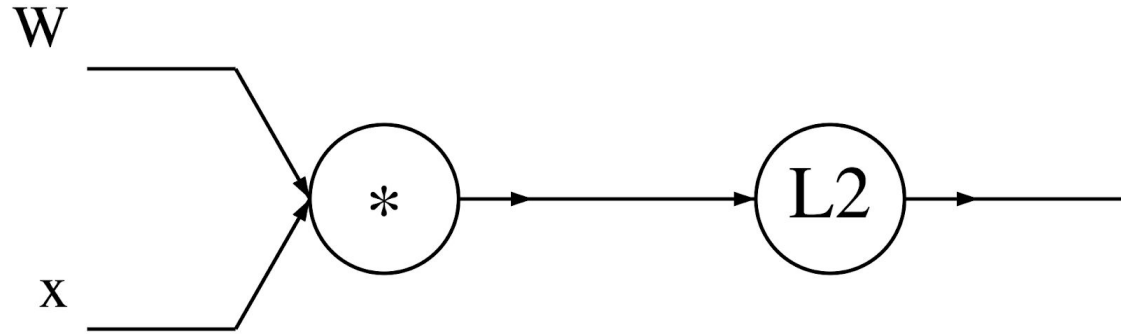


Diagram illustrating the domains of the variables x and W in the vectorized example:

$$\begin{array}{ccc} & \downarrow & \downarrow \\ & \in \mathbb{R}^n & \in \mathbb{R}^{n \times n} \end{array}$$

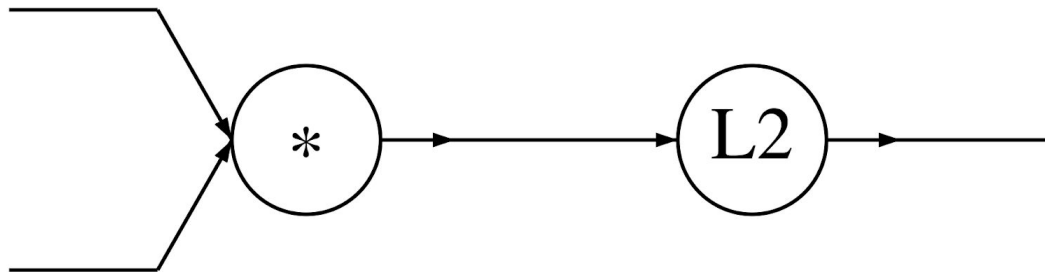
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$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix} \mathbf{W}$$

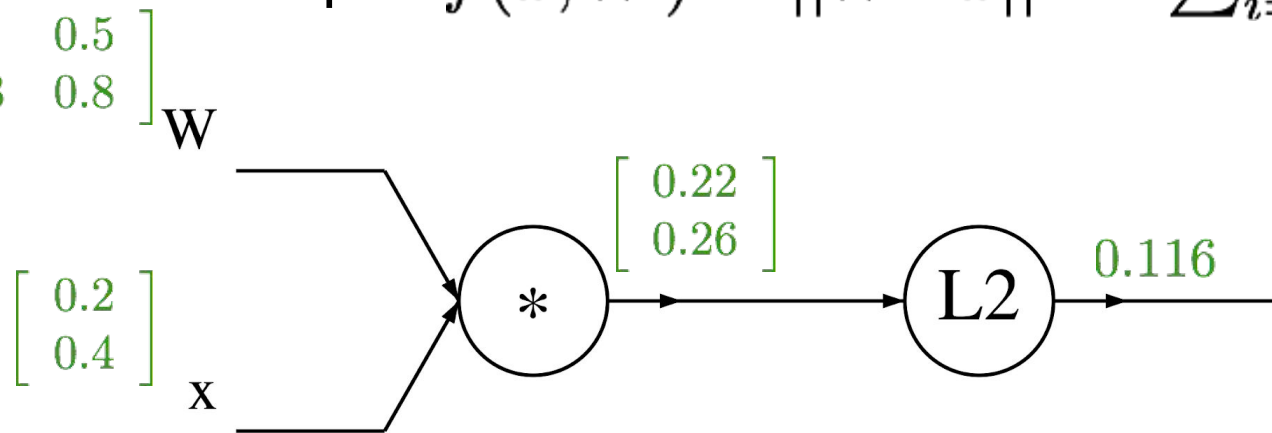
$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} \mathbf{x}$$



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

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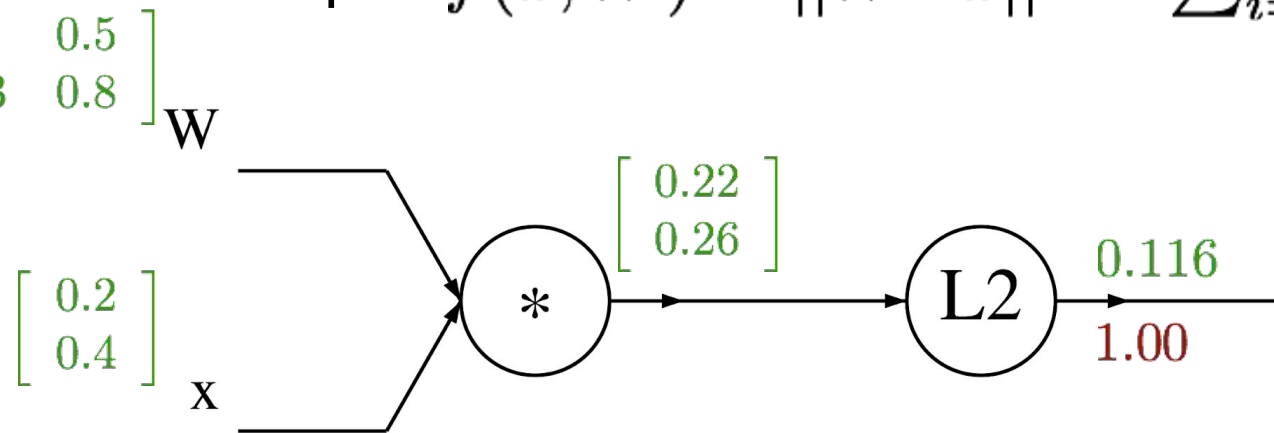
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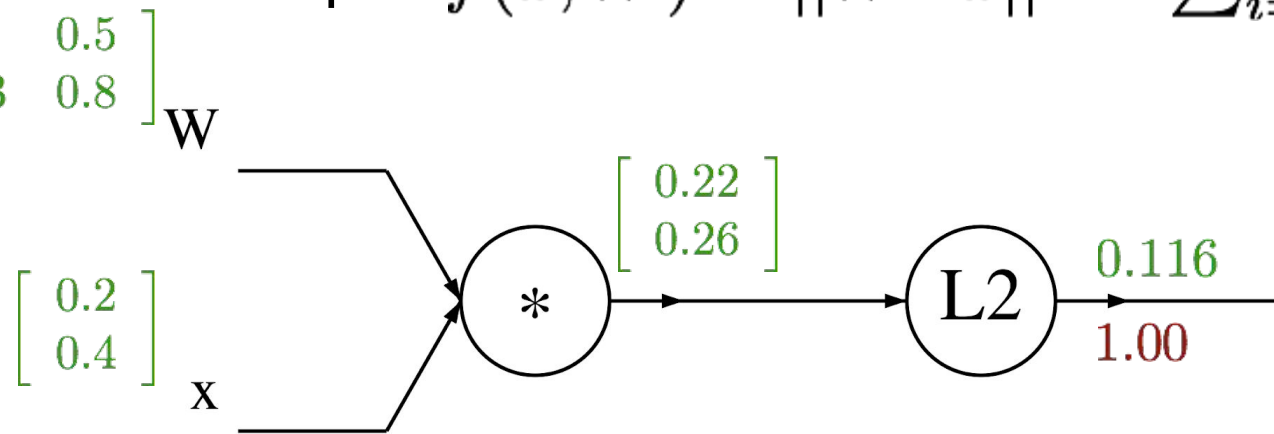
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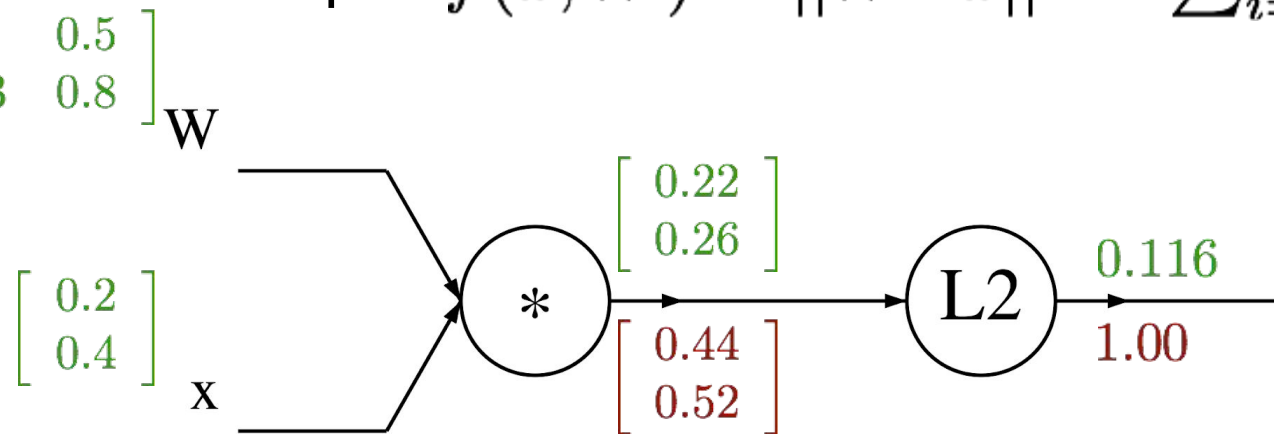
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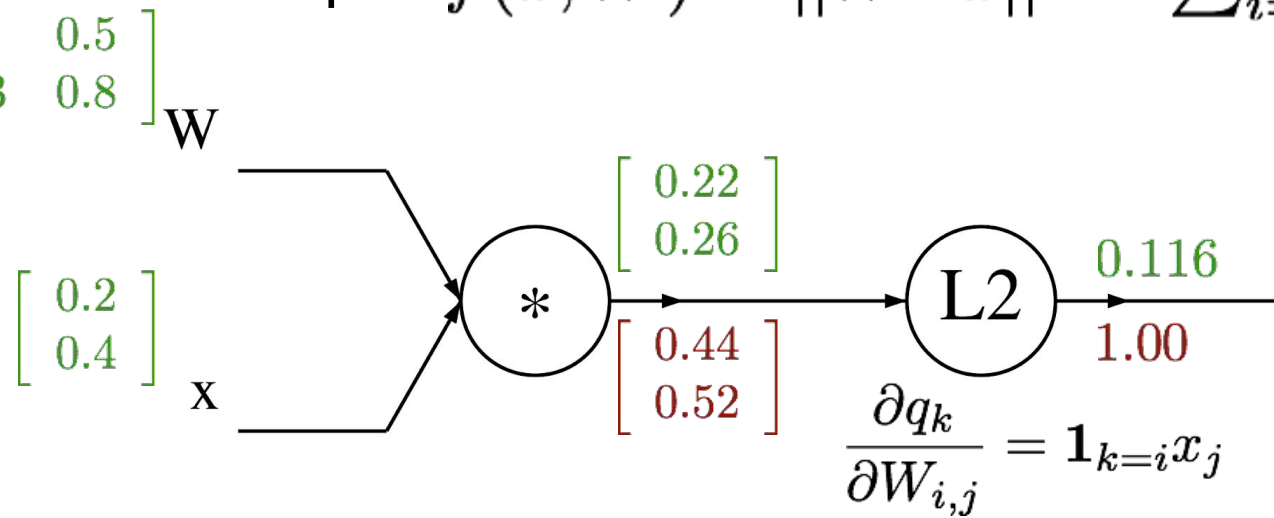
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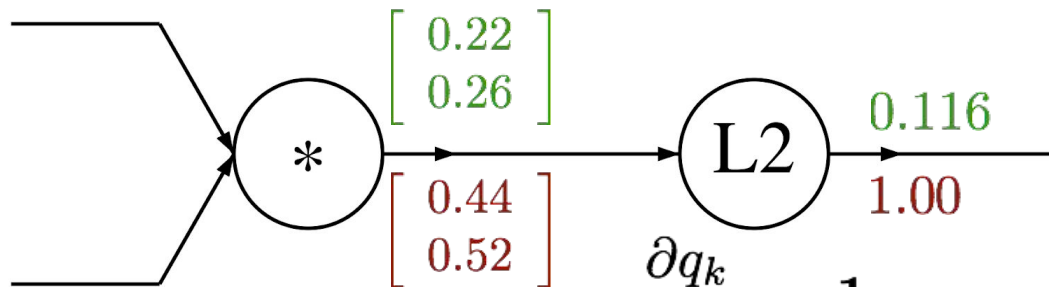
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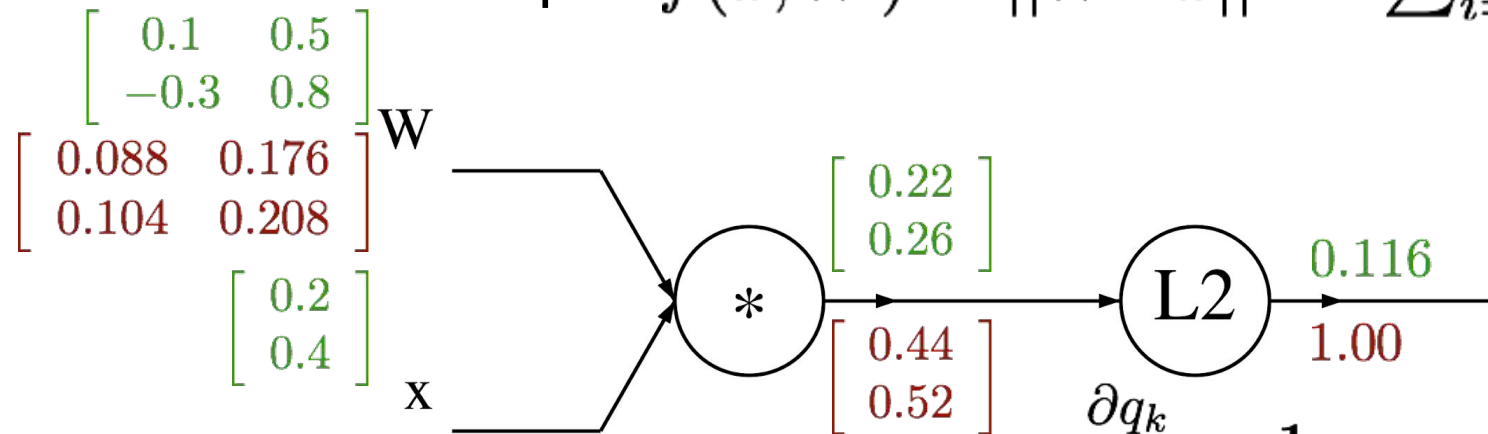
$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

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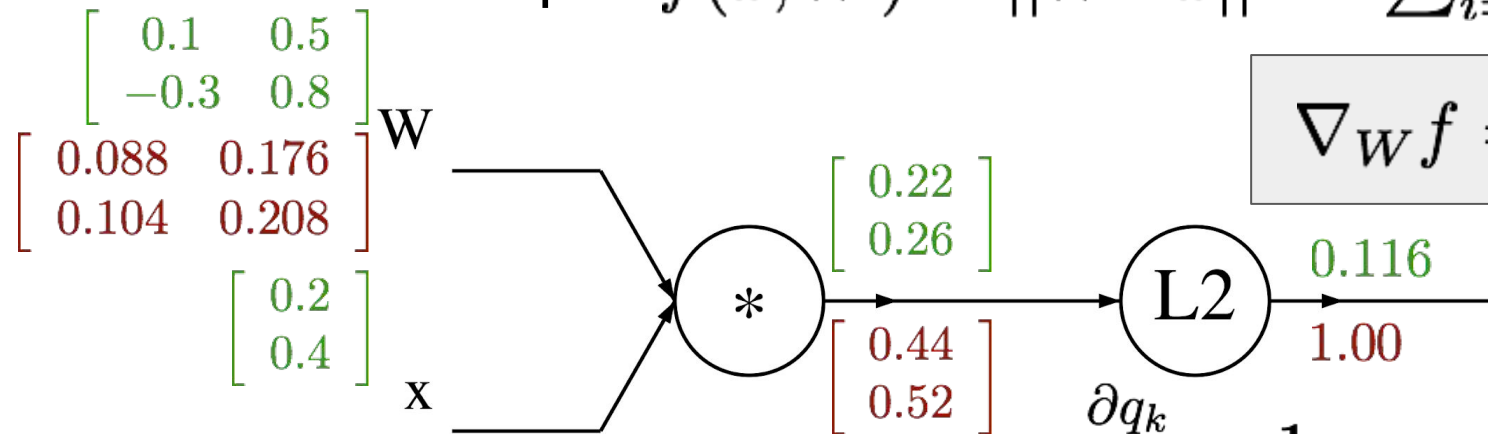


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$$\nabla_W f = 2q \cdot x^T$$

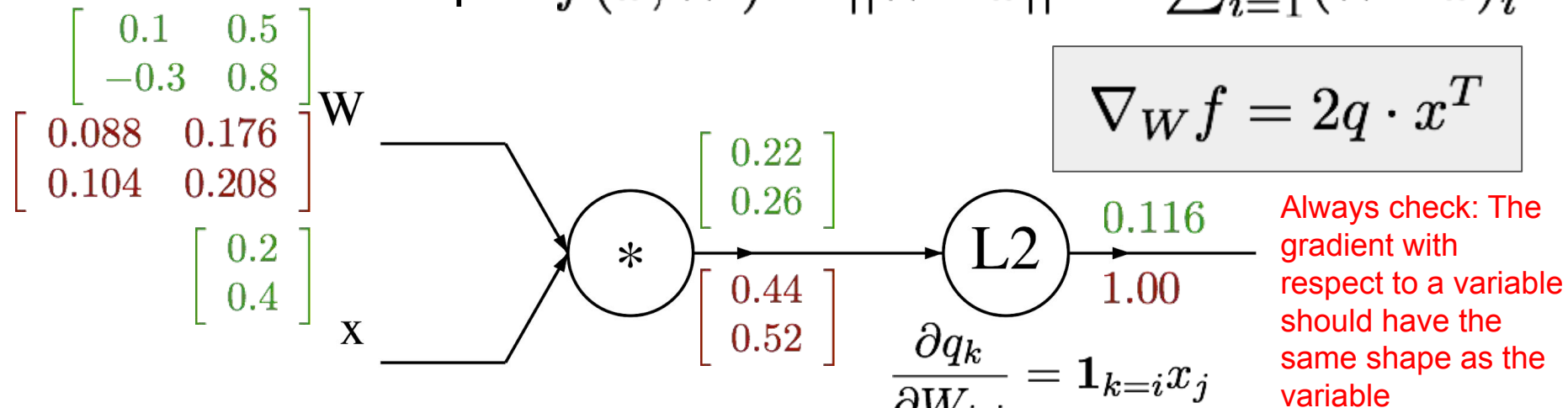
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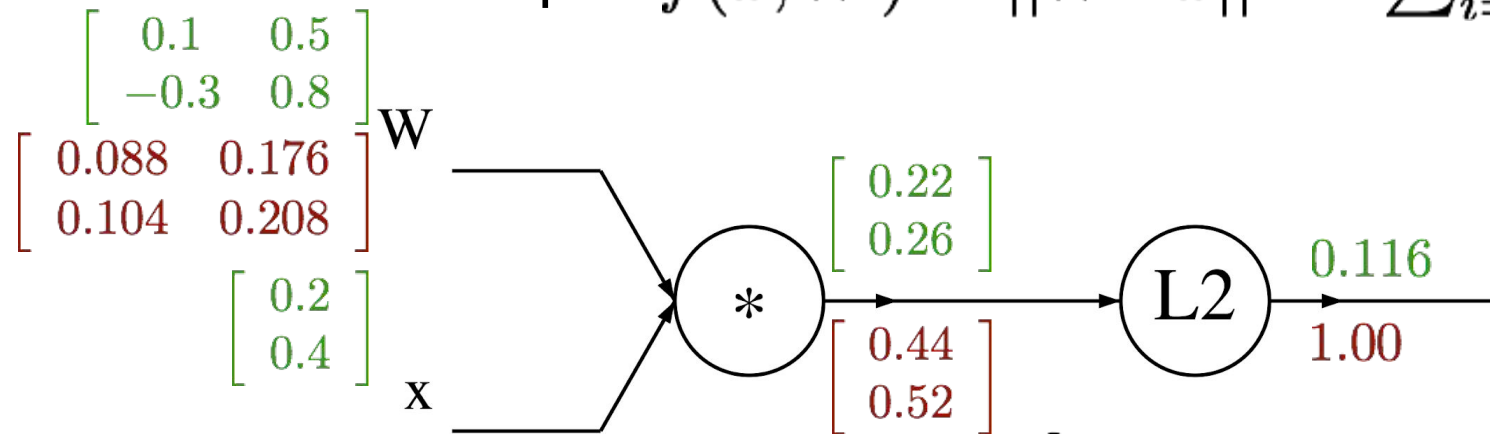


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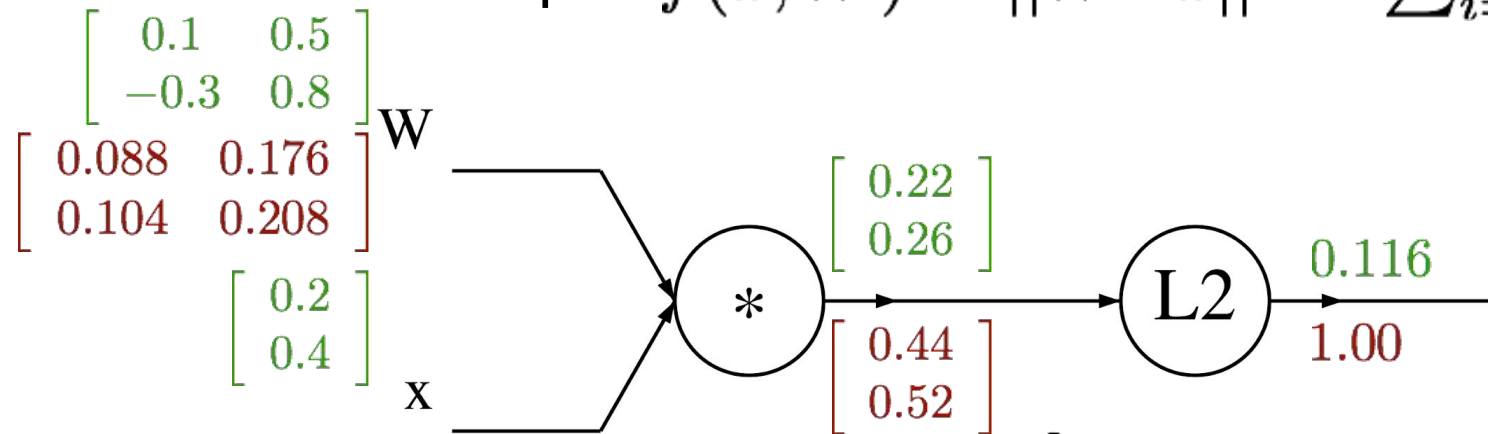


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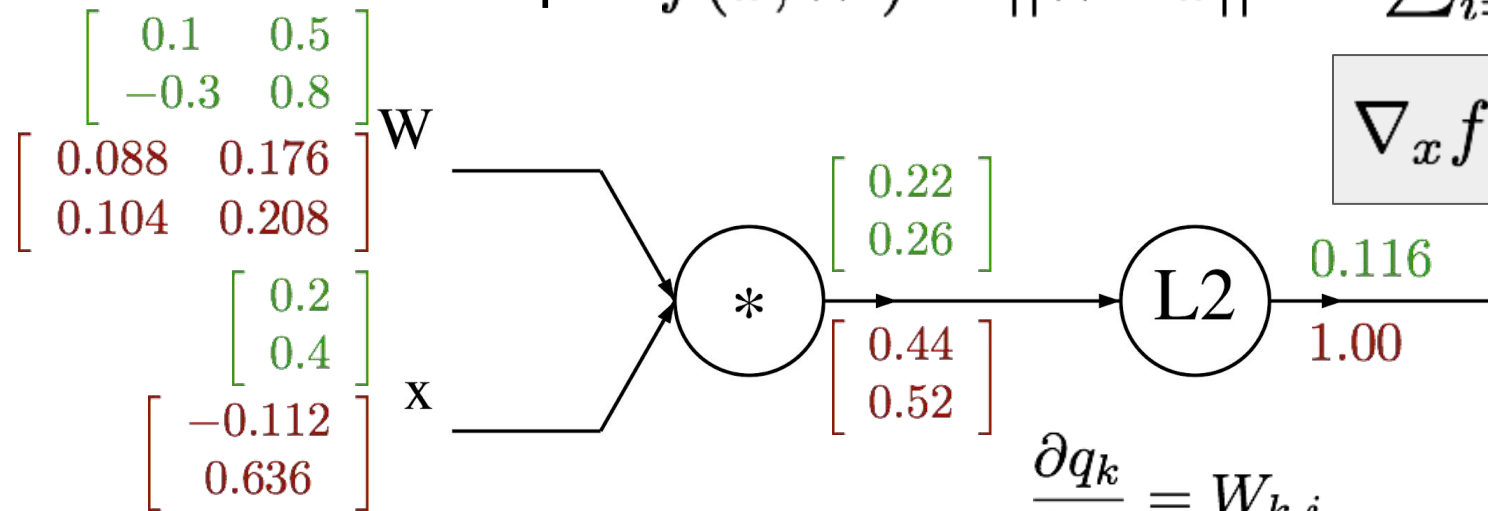


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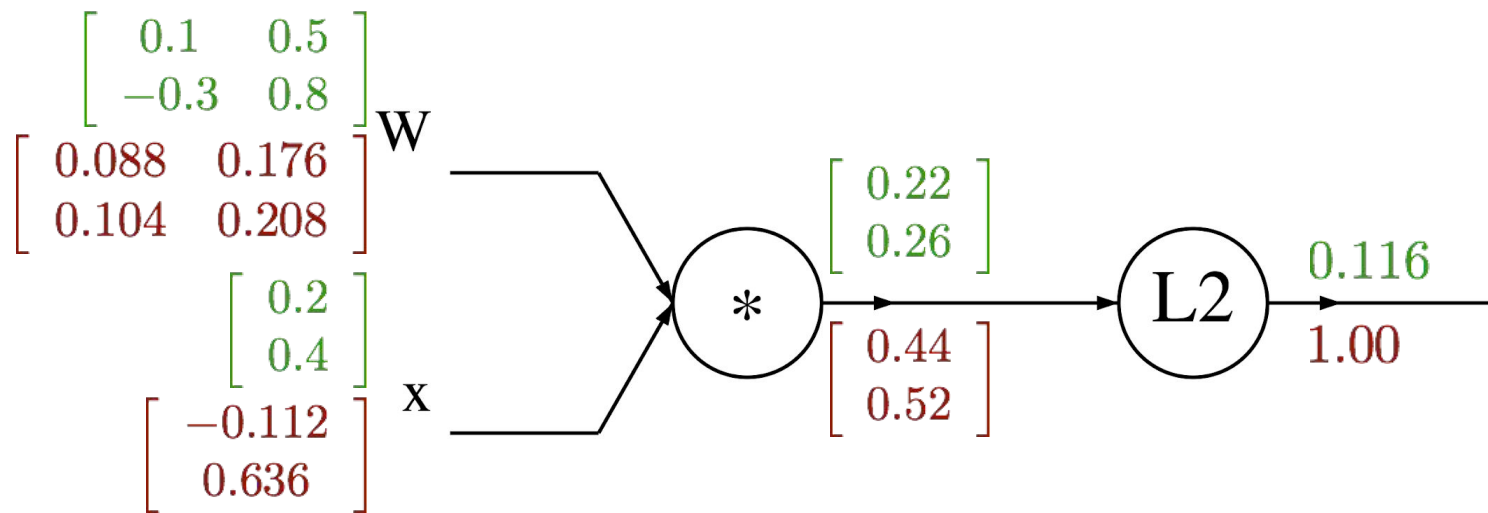


$$\nabla_x f = 2W^T \cdot q$$

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So far: backprop with scalars

Next time: vector-valued functions!

Next Time: Convolutional neural networks

